PROBLEM 37

Particle on a Spinning Elliptic Cone

Original statement · Problems.pdf, p. 25

Rotating Constraint Surface

A heavy point particle P with mass m moves without friction on a cone of elliptic cross-section as shown in the figure. We define an auxiliary reference frame $S_1 : \{O; \mathcal{B}_1\}$ attached to the cone. The cone is defined in terms of two parameters ζ, β as

$$\text{Elliptic cone:} \quad \begin{cases} x_1 = \zeta \cos\beta \\ y_1 = 2\zeta \sin\beta \\ z_1 = \zeta \end{cases} \qquad (5)$$

The cone spins about its Oz_1 axis at a constant angular rate Ω with respect to the inertial reference frame $S_0 : \{O; \mathcal{B}_0\}$. At time $t = 0$, the reference frame S_1 coincides with S_0, the particle P is located at $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$, and has a velocity $\mathbf{v}^P_{1,0} = \mathbf{i}_1$.

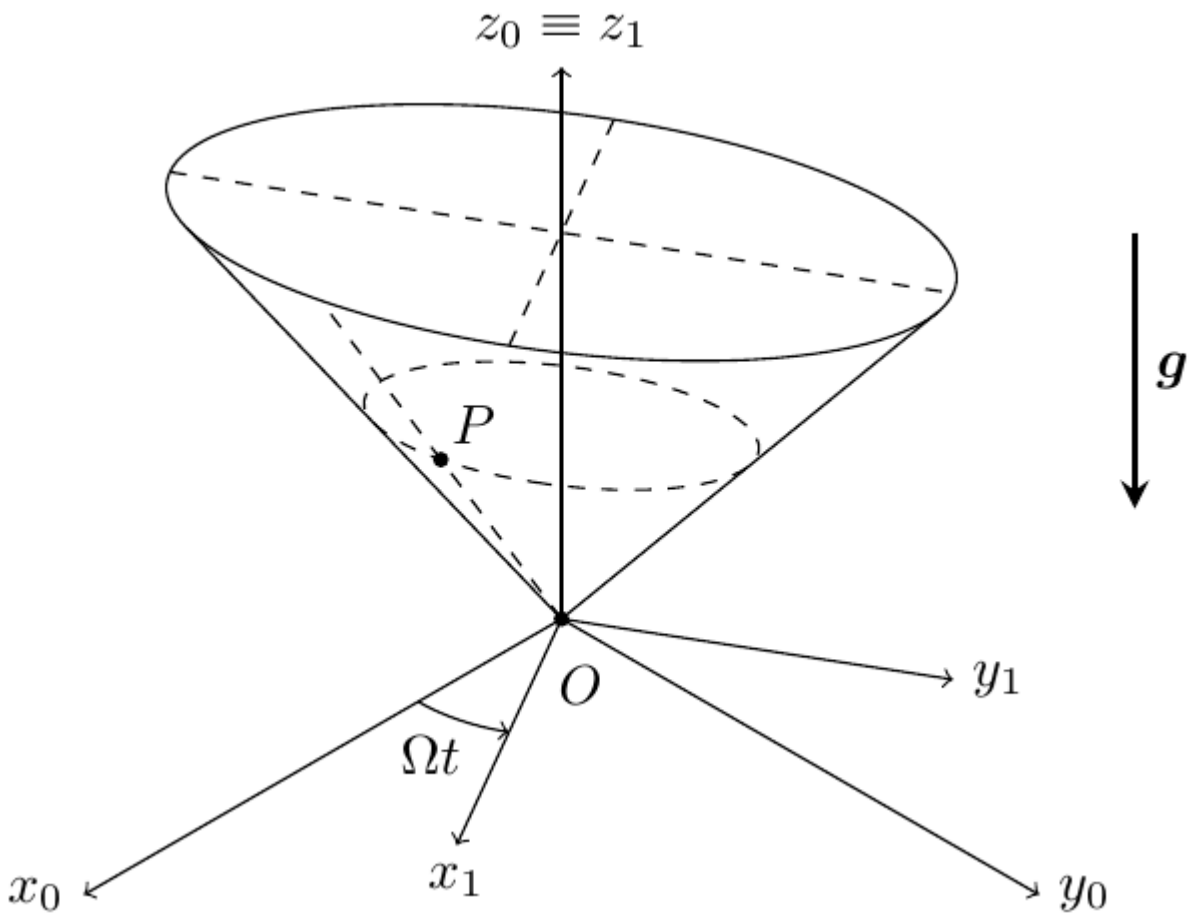


Figure: Elliptic cone with vertex O and vertical axis z0 = z1, frame S1 rotated by the angle Omega t about z0, particle P on the cone, gravity g downward (Problems.pdf, p. 25).

(a) How many degrees of freedom does the particle motion have?

(b) Obtain the general expression for the position, velocity, and acceleration of P **in reference frame** S_1 as a function of ζ, β and their derivatives. Remember to define any unit vectors that you use in terms of the vectors of $\mathcal{B}_0$.

(c) Identify and express mathematically all the forces acting on P as seen from S_1.

(d) Obtain two second-order differential equations for the evolution of ζ, β in time, which do not depend on any other unknowns in the problem.

(e)

Find, justify, and write down one conservation law in this problem, including the value of any constants there might be.

(f) Find the equilibrium positions for P in S_1 for $\zeta > 0$ as a function of the constants of the problem.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy point particle P of mass m slides on the inner surface of an elliptic cone with vertex O and vertical axis $Oz_1 \equiv Oz_0$. The cone spins about that axis with constant angular rate Ω relative to the inertial frame S_0.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; gravity is $\mathbf{g} = -g\,\mathbf{k}_0$.
  2. The contact is smooth (no friction) and bilateral: the reaction of the cone is normal to its surface.
  3. Ω is constant and the vertex O is fixed, so O is a common origin of S_0 and S_1.
  4. The data $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$ and $\mathbf{v}^P_{1,0} = \mathbf{i}_1$ are read in SI units (m and m/s).
- **Why a moving frame:** in S_0 the surface of the cone moves, so the constraint depends explicitly on time. In the frame S_1 attached to the cone the surface is fixed and is described by the two parameters (ζ, β) of Eq. (5). The price is that S_1 is not inertial, so Newton's law in S_1 needs the inertia forces of Notes Eqs. 4.62–4.63 (Slides 27–28); they are known functions of the relative position and velocity.

Symbol	Meaning	SI unit
m	mass of P	kg
g	gravitational acceleration	m/s^2
Ω	constant spin rate of the cone	rad/s
ζ	height of P above the vertex ($z_1 = \zeta$)	m
β	angular parameter of the elliptic section	rad

Symbol	Meaning	SI unit
N	magnitude of the normal reaction	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = \{O; \mathcal{B}_0\}$, $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with $\mathbf{k}_0$ vertical and upward.
- **Cone frame:** $S_1 = \{O; \mathcal{B}_1\}$, $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, attached to the cone. It is obtained from S_0 by a simple rotation of angle Ωt about $\mathbf{k}_0$ (the figure measures Ωt from x_0 to x_1), and coincides with S_0 at $t = 0$:

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

- **Change-of-basis matrix** (columns are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **Checks:** $\det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1$. The rows have unit norm ($\cos^2\Omega t + \sin^2\Omega t = 1$, and 1 for the third) and are mutually orthogonal ($-\cos\Omega t\sin\Omega t + \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.
- **Motion of S_1:** the origin is fixed, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. For a simple rotation about the fixed vector $\mathbf{k}_0$ (Slide 16), $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$. Check with the chain rule: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dfrac{d\cos\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{i}_0 + \dfrac{d\sin\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{j}_0 = \Omega\,(-\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0) = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$. Since Ω and $\mathbf{k}_1 = \mathbf{k}_0$ are constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom

Pedagogical Justification: a free particle has 3 coordinates; each independent holonomic constraint removes one.

Eliminating the parameters of Eq. (5): $\cos\beta = x_1/\zeta$, $\sin\beta = y_1/(2\zeta)$ and $\zeta = z_1$, so $\cos^2\beta + \sin^2\beta = 1$ gives the single constraint equation of the surface, fixed in S_1:

$$\frac{x_1^2}{z_1^2} + \frac{y_1^2}{4z_1^2} = 1 \iff x_1^2 + \frac{y_1^2}{4} - z_1^2 = 0$$

Hence $\mathrm{DOF} = 3 - 1 = 2$, and (ζ, β) are suitable generalized coordinates. Each horizontal section $z_1 = \zeta$ is an ellipse with semi-axes ζ (along $\mathbf{i}_1$) and 2ζ (along $\mathbf{j}_1$).

(b) Position, velocity and acceleration in S_1

Pedagogical Justification: O is the origin of S_1, so $\mathbf{r}_1^P = \overrightarrow{OP}$ (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$). The relative derivatives $\frac{d}{dt}\big|_1$ treat $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ as constants, so only the components are differentiated (product and chain rules).

$$\mathbf{r}_1^P = \zeta\cos\beta\,\mathbf{i}_1 + 2\zeta\sin\beta\,\mathbf{j}_1 + \zeta\,\mathbf{k}_1$$

Component by component, with $\frac{d}{dt}\cos\beta = \frac{d\cos\beta}{d\beta}\dot\beta = -\dot\beta\sin\beta$ and $\frac{d}{dt}\sin\beta = \dot\beta\cos\beta$:

$$\frac{d}{dt}(\zeta\cos\beta) = \dot\zeta\cos\beta - \zeta\dot\beta\sin\beta, \qquad \frac{d}{dt}(2\zeta\sin\beta) = 2\dot\zeta\sin\beta + 2\zeta\dot\beta\cos\beta, \qquad \frac{d}{dt}\zeta = \dot\zeta$$

$$\mathbf{v}_1^P = \Big(\dot\zeta\cos\beta - \zeta\dot\beta\sin\beta\Big)\,\mathbf{i}_1 + 2\Big(\dot\zeta\sin\beta + \zeta\dot\beta\cos\beta\Big)\,\mathbf{j}_1 + \dot\zeta\,\mathbf{k}_1$$

Differentiating again. For the $\mathbf{i}_1$ component, $\frac{d}{dt}(\dot\zeta\cos\beta) = \ddot\zeta\cos\beta - \dot\zeta\dot\beta\sin\beta$ and $\frac{d}{dt}(\zeta\dot\beta\sin\beta) = \dot\zeta\dot\beta\sin\beta + \zeta\ddot\beta\sin\beta + \zeta\dot\beta^2\cos\beta$; for the $\mathbf{j}_1$ component, $\frac{d}{dt}(\dot\zeta\sin\beta) = \ddot\zeta\sin\beta + \dot\zeta\dot\beta\cos\beta$ and $\frac{d}{dt}(\zeta\dot\beta\cos\beta) = \dot\zeta\dot\beta\cos\beta + \zeta\ddot\beta\cos\beta - \zeta\dot\beta^2\sin\beta$. Collecting terms:

$$\mathbf{a}_1^P = \left[\left(\ddot\zeta - \zeta\dot\beta^2\right)\cos\beta - \left(2\dot\zeta\dot\beta + \zeta\ddot\beta\right)\sin\beta\right]\mathbf{i}_1 + 2\left[\left(\ddot\zeta - \zeta\dot\beta^2\right)\sin\beta + \left(2\dot\zeta\dot\beta + \zeta\ddot\beta\right)\cos\beta\right]\mathbf{j}_1 + \ddot\zeta\,\mathbf{k}_1$$

The unit vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are defined in terms of $\mathcal{B}_0$ in Phase 2. For later use we write $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$ and $\mathbf{v}_1^P = \dot x_1\mathbf{i}_1 + \dot y_1\mathbf{j}_1 + \dot z_1\mathbf{k}_1$ with the components above.

(c) Forces acting on P as seen from S_1

> **Pedagogical Justification:** in the non-inertial frame S_1 Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), where $\mathbf{F}$ are the actual forces and $\mathbf{F}_i$ the inertia forces of Notes Eq. 4.62 (Slide 28).

1. Weight: $\mathbf{W} = -mg\,\mathbf{k}_0 = -mg\,\mathbf{k}_1$.

2. Normal reaction. Two tangent vectors to the cone at P are the partial derivatives of $\mathbf{r}_1^P$ (they are not unit vectors):

$$\mathbf{e}_\zeta \equiv \frac{\partial \mathbf{r}_1^P}{\partial \zeta} = \cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1 + \mathbf{k}_1, \qquad \frac{\partial \mathbf{r}_1^P}{\partial \beta} = \zeta\,\mathbf{e}_\beta, \quad \mathbf{e}_\beta \equiv -\sin\beta\,\mathbf{i}_1 + 2\cos\beta\,\mathbf{j}_1$$

A normal vector is their cross product, expanded component by component:

$$\mathbf{e}_\zeta \times \mathbf{e}_\beta = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ \cos\beta & 2\sin\beta & 1 \\ -\sin\beta & 2\cos\beta & 0 \end{vmatrix} = (0 - 2\cos\beta)\,\mathbf{i}_1 - (0 + \sin\beta)\,\mathbf{j}_1 + \left(2\cos^2\beta + 2\sin^2\beta\right)\mathbf{k}_1 = -2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1$$

Its squared norm is $4\cos^2\beta + \sin^2\beta + 4 = 5 + 3\cos^2\beta$, so the smooth reaction is

$$\mathbf{N} = N\,\mathbf{n}, \qquad \mathbf{n} = \frac{-2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1}{\sqrt{5 + 3\cos^2\beta}}$$

with N an unknown scalar. (Check: the gradient of $x_1^2 + y_1^2/4 - z_1^2$ is $(2x_1, y_1/2, -2z_1) = \zeta(2\cos\beta, \sin\beta, -2)$, parallel to $\mathbf{n}$.)

3. Inertia forces (Notes Eq. 4.62). Since $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, the force due to the acceleration of the origin and the Euler force vanish: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$, $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.

Centrifugal force. With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{k}_1 \times (x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1) = \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1)$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \Omega\,\mathbf{k}_1 \times \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1) = -\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1)$$

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = m\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1) = m\Omega^2\zeta\,(\cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1)$$

which is $m\Omega^2\mathbf{r}_\perp$, radially outward from the axis (Slide 30).

Coriolis force.

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = -2m\Omega\,\mathbf{k}_1 \times (\dot{x}_1\mathbf{i}_1 + \dot{y}_1\mathbf{j}_1 + \dot{z}_1\mathbf{k}_1) = 2m\Omega\,(\dot{y}_1\,\mathbf{i}_1 - \dot{x}_1\,\mathbf{j}_1)$$

$$\mathbf{F}_{COR} = 4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\mathbf{i}_1 - 2m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\mathbf{j}_1$$

Equation of motion in S_1 (Notes Eq. 4.63):

$$m\mathbf{a}_1^P = -mg\,\mathbf{k}_1 + N\,\mathbf{n} + \mathbf{F}_C + \mathbf{F}_{COR}$$

(d) Two equations of motion free of the reaction

> **Pedagogical Justification:** the only unknown besides ζ and β is N, and $\mathbf{N}$ is perpendicular to the surface. Projecting the vector equation on the two tangent vectors $\mathbf{e}_\zeta$ and $\mathbf{e}_\beta$ eliminates it: $(-2\cos\beta)\cos\beta + (-\sin\beta)(2\sin\beta) + 2 = 0$ and $(-2\cos\beta)(-\sin\beta) + (-\sin\beta)(2\cos\beta) = 0$.

Projection on $\mathbf{e}_\zeta = (\cos\beta, 2\sin\beta, 1)$. Left-hand side:

$$\mathbf{a}_1^P \cdot \mathbf{e}_\zeta = \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(\cos^2\beta + 4\sin^2\beta\right) + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(-\sin\beta\cos\beta + 4\sin\beta\cos\beta\right) + \ddot{\zeta}$$

$$= \ddot{\zeta} + \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(1 + 3\sin^2\beta\right) + 3\left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\sin\beta\cos\beta$$

Right-hand side, term by term: $-mg\,\mathbf{k}_1 \cdot \mathbf{e}_\zeta = -mg$; $\mathbf{F}_C \cdot \mathbf{e}_\zeta = m\Omega^2\zeta\left(\cos^2\beta + 4\sin^2\beta\right) = m\Omega^2\zeta\left(1 + 3\sin^2\beta\right)$;

$$\mathbf{F}_{COR} \cdot \mathbf{e}_\zeta = 4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\cos\beta - 4m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\sin\beta = 4m\Omega\zeta\dot{\beta}\left(\cos^2\beta + \sin^2\beta\right) = 4m\Omega\zeta\dot{\beta}$$

Dividing by m:

$$\boxed{\ddot{\zeta} + \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(1 + 3\sin^2\beta\right) + 3\left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\cos\beta\sin\beta = -g + \Omega^2\zeta\left(1 + 3\sin^2\beta\right) + 4\Omega\zeta\dot{\beta}}$$

Projection on $\mathbf{e}_\beta = (-\sin\beta, 2\cos\beta, 0)$. Left-hand side:

$$\mathbf{a}_1^P \cdot \mathbf{e}_\beta = \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(-\sin\beta\cos\beta + 4\sin\beta\cos\beta\right) + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(\sin^2\beta + 4\cos^2\beta\right) = 3\left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\sin\beta\cos\beta + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(1 + 3\cos^2\beta\right)$$

Right-hand side: $-mg\,\mathbf{k}_1 \cdot \mathbf{e}_\beta = 0$; $\mathbf{F}_C \cdot \mathbf{e}_\beta = m\Omega^2\zeta\left(-\cos\beta\sin\beta + 4\sin\beta\cos\beta\right) = 3m\Omega^2\zeta\sin\beta\cos\beta$;

$$\mathbf{F}_{COR}\cdot\mathbf{e}_\beta=-4m\Omega\left(\dot\zeta\sin\beta+\zeta\dot\beta\cos\beta\right)\sin\beta-4m\Omega\left(\dot\zeta\cos\beta-\zeta\dot\beta\sin\beta\right)\cos\beta=-4m\Omega\dot\zeta$$

Dividing by m:

$$\boxed{3\left(\ddot\zeta-\zeta\dot\beta^2\right)\cos\beta\sin\beta+\left(2\dot\zeta\dot\beta+\zeta\ddot\beta\right)\left(1+3\cos^2\beta\right)=3\Omega^2\zeta\cos\beta\sin\beta-4\Omega\dot\zeta}$$

These two equations contain only ζ, β and their derivatives. The initial conditions follow from the data: $\mathbf{r}^P_{1,0}=2\mathbf{j}_1+\mathbf{k}_1$ gives $\zeta_0=1$, $\zeta_0\cos\beta_0=0$ and $2\zeta_0\sin\beta_0=2$, so $\beta_0=\pi/2$; $\mathbf{v}^P_{1,0}=\mathbf{i}_1$ gives $\dot\zeta_0=0$ ($\mathbf{k}_1$ component) and $\dot\zeta_0\cos\beta_0-\zeta_0\dot\beta_0\sin\beta_0=-\dot\beta_0=1$, so $\dot\beta_0=-1$ rad/s; the $\mathbf{j}_1$ component $2(\dot\zeta_0\sin\beta_0+\zeta_0\dot\beta_0\cos\beta_0)=0$ is consistent (the initial velocity is tangent to the cone).

(e) Conservation law: mechanical energy in S_1

> **Pedagogical Justification:** E^P_1 is conserved when every force that does work in S_1, inertia forces included, derives from a potential (Slide 29). Here: $\mathbf{N}\cdot\mathbf{v}^P_1=0$ because the cone is fixed in S_1 and $\mathbf{v}^P_1$ is tangent to it; $\mathbf{F}_{COR}\cdot\mathbf{v}^P_1=0$ always (Slide 31); $\mathbf{F}_A=\mathbf{F}_E=\mathbf{0}$; the weight is a constant force in S_1 ($\mathbf{k}_1=\mathbf{k}_0$) with potential mgz_1; and, since $\boldsymbol{\omega}_{10}$ is constant, the centrifugal force has potential $-\frac{1}{2}m\Omega^2r_\perp^2$ (Slide 30).

Kinetic energy in S_1, expanding the squares of the components of $\mathbf{v}^P_1$:

$$\left|\mathbf{v}^P_1\right|^2=\left(\dot\zeta^2\cos^2\beta-2\zeta\dot\zeta\dot\beta\sin\beta\cos\beta+\zeta^2\dot\beta^2\sin^2\beta\right)+4\left(\dot\zeta^2\sin^2\beta+2\zeta\dot\zeta\dot\beta\sin\beta\cos\beta+\zeta^2\dot\beta^2\cos^2\beta\right)+\dot\zeta^2$$

$$T^P_1=\frac{m}{2}\left[\left(2+3\sin^2\beta\right)\dot\zeta^2+\left(1+3\cos^2\beta\right)\zeta^2\dot\beta^2+6\zeta\dot\zeta\dot\beta\sin\beta\cos\beta\right]$$

Potential in S_1, with $z_1=\zeta$ and $r_\perp^2=x_1^2+y_1^2=\zeta^2\left(\cos^2\beta+4\sin^2\beta\right)=\zeta^2\left(1+3\sin^2\beta\right)$:

$$V_1 = mg\zeta - \frac{1}{2}m\Omega^2\zeta^2\left(1+3\sin^2\beta\right)$$

Direct check that $E_1^P = T_1^P + V_1$ is constant. Since $\mathbf{v}_1^P = \dot\zeta\,\mathbf{e}_\zeta + \zeta\dot\beta\,\mathbf{e}_\beta$ and $\dfrac{dV_1}{dt} = -\left(\mathbf{W}+\mathbf{F}_C\right)\cdot\mathbf{v}_1^P$ (chain rule on $V_1(\mathbf{r}_1^P(t))$):

$$\frac{dE_1^P}{dt} = \left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{v}_1^P = \dot\zeta\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\zeta + \zeta\dot\beta\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\beta$$

and by the two equations of part (d) the brackets equal $4m\Omega\zeta\dot\beta$ and $-4m\Omega\dot\zeta$, so

$$\frac{dE_1^P}{dt} = \dot\zeta\left(4m\Omega\zeta\dot\beta\right) + \zeta\dot\beta\left(-4m\Omega\dot\zeta\right) = 0$$

Conservation law:

$$\boxed{\frac{m}{2}\left[\left(2+3\sin^2\beta\right)\dot\zeta^2 + \left(1+3\cos^2\beta\right)\zeta^2\dot\beta^2 + 6\zeta\dot\zeta\dot\beta\sin\beta\cos\beta\right] + mg\zeta - \frac{m}{2}\Omega^2\zeta^2\left(1+3\sin^2\beta\right) = E_1^P = m\left(\frac{1}{2}+g-2\Omega^2\right)}$$

The constant comes from $t=0$: $\zeta_0 = 1$ m, $\beta_0 = \pi/2$, $\dot\zeta_0 = 0$, $\dot\beta_0 = -1$ rad/s give $T_1^P(0) = \frac{m}{2}(1+0)(1)^2(-1)^2 = \frac{m}{2}$ (that is, $\frac{1}{2}m|\mathbf{v}_{1,0}^P|^2$ with $|\mathbf{v}_{1,0}^P| = 1$ m/s) and $V_1(0) = mg(1) - \frac{1}{2}m\Omega^2(1)^2(1+3) = m\left(g-2\Omega^2\right)$, in joules when m, g and Ω are in SI units.

The angular momentum about Oz_0 is **not** conserved: the moment of $\mathbf{N}$ about the axis is $\left(\mathbf{r}_1^P\times\mathbf{n}\right)\cdot\mathbf{k}_1\,N = (x_1 n_y - y_1 n_x)\,N = \dfrac{\zeta\left(-\cos\beta\sin\beta + 4\sin\beta\cos\beta\right)}{\sqrt{5+3\cos^2\beta}}N = \dfrac{3\zeta\sin\beta\cos\beta}{\sqrt{5+3\cos^2\beta}}N$, which does not vanish in general because the section is not circular.

(f) Equilibrium positions in S_1

> **Pedagogical Justification:** an equilibrium in S_1 is a solution with ζ and β constant, so $\dot\zeta = \dot\beta = \ddot\zeta = \ddot\beta = 0$ in the two equations of part (d); the Coriolis terms vanish automatically.

The $\mathbf{e}_\beta$ equation reduces to $0 = 3\Omega^2\zeta\cos\beta\sin\beta = \frac{3}{2}\Omega^2\zeta\sin 2\beta$. With $\zeta > 0$ and $\Omega \neq 0$, $\sin 2\beta = 0$:

$$\beta \in \left\{0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}\right\}$$

The $\mathbf{e}_\zeta$ equation reduces to $0 = -g + \Omega^2\zeta\left(1 + 3\sin^2\beta\right)$, so

$$\zeta_{eq} = \frac{g}{\Omega^2\left(1 + 3\sin^2\beta\right)}$$

β	ζ_{eq}	**Position** $\mathbf{r}_1^P$
0	g/Ω^2	$\frac{g}{\Omega^2}(\mathbf{i}_1 + \mathbf{k}_1)$
π	g/Ω^2	$\frac{g}{\Omega^2}(-\mathbf{i}_1 + \mathbf{k}_1)$
$\pi/2$	$g/(4\Omega^2)$	$\frac{g}{2\Omega^2}\mathbf{j}_1 + \frac{g}{4\Omega^2}\mathbf{k}_1$
$3\pi/2$	$g/(4\Omega^2)$	$-\frac{g}{2\Omega^2}\mathbf{j}_1 + \frac{g}{4\Omega^2}\mathbf{k}_1$

Force-balance check at $\beta = 0$: $\mathbf{n} = (-2, 0, 2)/\sqrt{8}$, so $\mathbf{0} = -mg\,\mathbf{k}_1 + m\Omega^2\zeta\,\mathbf{i}_1 + \frac{N}{\sqrt{2}}(-\mathbf{i}_1 + \mathbf{k}_1)$ gives $N = \sqrt{2}\,mg$ and $\Omega^2\zeta = g$. **At $\beta = \pi/2$:** $\mathbf{n} = (0, -1, 2)/\sqrt{5}$, so $\mathbf{0} = -mg\,\mathbf{k}_1 + 2m\Omega^2\zeta\,\mathbf{j}_1 + \frac{N}{\sqrt{5}}(-\mathbf{j}_1 + 2\mathbf{k}_1)$ gives $N = \frac{\sqrt{5}}{2}mg$ and $2\Omega^2\zeta = \frac{g}{2}$, i.e. $\zeta = g/(4\Omega^2)$. Both agree with the table.

Extension (stability, not asked): the Hessian of $V_1(\zeta, \beta)$ is $\mathrm{diag}\left(-m\Omega^2,\ -3mg^2/\Omega^2\right)$ at $\beta = 0, \pi$ and $\mathrm{diag}\left(-4m\Omega^2,\ 3mg^2/(16\Omega^2)\right)$ at $\beta = \pi/2, 3\pi/2$. V_1 has no minimum (along each generator it is a downward parabola in ζ), so the energy criterion cannot certify the stability of any of these equilibria.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Meaning of the equilibria:** they lie on the four generators contained in the principal planes x_1z_1 and y_1z_1 of the cone. Only there does the normal $\mathbf{n}$ lie in the vertical plane through the axis, so that it can balance the weight and the (radial) centrifugal force without a tangential remainder. On the steeper generators ($\beta = 0, \pi$, slope 45°) the particle must be higher, $\zeta = g/\Omega^2$, than on the flatter ones ($\beta = \pm\pi/2$, $\zeta = g/(4\Omega^2)$).
- **Gyroscopic coupling:** the Coriolis terms $4\Omega\zeta\dot{\beta}$ and $-4\Omega\dot{\zeta}$ couple the two equations antisymmetrically, which is why they cancel in dE_1^P/dt (Coriolis never does work).

- **Limit $\Omega \to 0$:** the centrifugal and Coriolis terms vanish, E_1^P becomes the ordinary energy $\frac{1}{2}mv^2 + mgz$, and $\zeta_{eq} \to \infty$: a fixed, vertex-down smooth cone has no equilibrium position with $\zeta > 0$.
- **Dimensions:** in the $\mathbf{e}_\zeta$ equation, $[\ddot{\zeta}] = [\zeta\dot{\beta}^2] = [\dot{\zeta}\dot{\beta}] = [\zeta\ddot{\beta}] = \mathrm{m/s^2}$, $[g] = \mathrm{m/s^2}$, $[\Omega^2\zeta] = \mathrm{s^{-2}\,m}$, $[\Omega\zeta\dot{\beta}] = \mathrm{s^{-1}\,m\,s^{-1}}$; the $\mathbf{e}_\beta$ equation has the same terms. $[E_1^P] = \mathrm{kg\,m^2/s^2} = \mathrm{J}$; $[g/\Omega^2] = \mathrm{(m/s^2)/s^{-2}} = \mathrm{m}$.
- **Official key:** both equations of part (d) coincide with the official key (Problems.pdf, p. 75) term by term.
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals $\mathbf{a}_1^P + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ (Notes Eq. 3.27); both projections match the key; $dE_1^P/dt = 0$ along the equations of motion; the four equilibria satisfy both equations.

Source: Problems.pdf (PDF page 25); key on PDF page 75